Zicheng Xie

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EDUCATION

The Chinese University of Hong Kong (CUHK), Hong Kong 08/2026 - 07/2030 (expected)
*Ph.D. in **Statistics***

- **Postgraduate Studentship Award**

Southern University of Science and Technology (SUSTech), Shenzhen, China 08/2022 - 06/2026 (expected)
*Bachelor of Science in **Data Science and Big Data Technology***

- **Cumulative GPA:** 3.91/4.0 | **Rank:** 2/49
- **National Encouragement Scholarship**, 2024, 2025
- **First-class Scholarship** for Outstanding Student (top 5%), 2023, 2025

Nanyang Technological University (NTU), Singapore 01/2025 - 05/2025
*Exchange Program in **Mathematical and Computer Sciences***

- **Relevant Coursework:** Real Analysis II (Measure-based), Basic Optimization, Financial Econometrics.

University of Oxford, Oxford, UK 07/2023 - 08/2023
*Summer School in **Artificial Intelligence and Machine Learning***

- **Project:** Fine-grained image classification based on Recurrent Attention Convolutional Neural Network.

MANUSCRIPT & PATENT

[1] Zhengdong Huang[†], **Zicheng Xie**[†], Wentao Tian, Jingyu Liu, Lunhong Dong*, Peng Yang*. *Learning Representations from Heterogeneous Data for Robust Heart Rate Modeling.* arXiv:2508.21785 [cs.LG], 2025. doi:10.48550/arXiv.2508.21785. (*Co-first author*)

[2] Jingyu Liu, Zhengdong Huang, **Zicheng Xie**, Wentao Tian, Zihan Li, Zihao Chen, Lunhong Dong, Chengyu Huang, Mengzhe Du. Middle- and Long-Distance Running Training Management and Data Analysis System. Chinese patent application, published as CN120655464 A.

[3] **Zicheng Xie**, Jingyu Liu, Zhengdong Huang, et al. Method, System, Terminal, and Medium for Training Plan Generation Based on Large-Language-Model Multi-Agent Collaboration. Chinese patent application, Application No. 202610438142.9, filed Apr 3, 2026.

RESEARCH EXPERIENCE

ParroTao: AI Platform for Personalized Training Analytics and Program Generation 09/2024 - Present
Project Lead, Supervisor: Prof. Peng Yang, SUSTech

- Developed an AI platform for personalized sports analytics using heterogeneous wearable data, integrating multi-device workout streams, physiological signals, environmental variables, and user metadata for training evaluation and decision support.
- Designed a representation learning framework for robust heart rate prediction under cross-device and cross-user heterogeneity, combining random feature dropout, time-aware attention, and contrastive learning.
- Built and benchmarked ParroTao, a large-scale multi-device, multi-activity wearable dataset for modeling real-world heterogeneity in sports data.
- Developed a self-evolving multi-agent LLM pipeline for personalized training plan generation, using structured task context and domain-rule retrieval to iteratively generate, evaluate, and refine training plans.
- ***Achievements:***
  (1) Co-first-authored manuscript *"Learning Representations from Heterogeneous Data for Robust Heart Rate Modeling"*, submitted to IEEE J-BHI.
  (2) One patent on AI-driven training management and data analytics for middle- and long-distance running, and one additional patent application on multi-agent LLM-based training plan generation.
  (3) Project supported by Guangdong Province "Climbing Program" Special Funds (Grant No. 2025C12601 and 2026C12601; CNY 40,000) and by Huawei Health Kit and Garmin Developer Program.

Multi-modal Topic Modeling for Single-cell Perturb-seq Data 09/2025 - Present

Supervisor: Prof. Yingxin Lin, CUHK

- Develop multimodal representation learning methods for large-scale single-cell Perturb-seq data to identify biologically meaningful latent programs and capture regulator–program–trait relationships.
- Integrate transcriptomic profiles, perturbation identities, gene–trait supervision, and pretrained foundation-model embeddings into a unified learning framework.
- Benchmark the framework across multiple single-cell datasets using downstream prediction, classification, and embedding quality evaluation tasks.
- Evaluate learned latent factors with diversity, enrichment, and interpretability analyses to assess robustness, transferability, and biological relevance.

Robust Multimodal Learning with Missing Modalities 01/2025 – 09/2025

Supervisor: Prof. Doudou Zhou, NUS

- Developed a pattern-aware, share-specific multimodal learning framework that addresses three key challenges: insufficient cross-modal integration, robustness to missing modalities, and distribution shifts in downstream tasks.
- Improved robustness and predictive accuracy using contrastive pretraining with random modality dropout and patient-level embedding decomposition.
- Designed a two-step estimation algorithm for downstream tasks, achieving bias correction and robust estimation in the downstream tasks.
- Validated the framework and achieved state-of-the-art performance on the MIMIC-IV dataset, outperforming existing methods under diverse missing-modality scenarios.

UWB/IMU-Fused Motion Capture System for 3D Human Center-of-Mass Estimation 10/2022 - 08/2023

Team Member, Supervisor: Prof. Mingming Zhang, SUSTech

- Designed a wearable motion capture system combining UWB communication and multi-IMU sensor fusion for real-time 3D CoM estimation, targeting applications in fall detection and balance assessment for elderly care.
- Integrated multi-IMU sensor fusion and UWB-IMU localization algorithms to achieve sub-centimeter accuracy and magnetic interference resilience.
- Built two hardware-software prototypes (WiFi/UWB-based) for clinical validation.
- ***Achievements:***
  (1) Project selected for the National College Student Innovation Program (CNY 20,000 funding).
  (2) Project supported by Guangdong Province "Climbing Program" Special Funds (Grant No. 2023C11305; CNY 20,000 funding).

RESEARCH SEMINAR

Talks @ Machine Learning Theory Seminar, SUSTech 07/2024 - 09/2024

Instructed by Prof. Hongxin Wei and Dr. Hao Zeng

- Participated in seminars discussing topics such as supervised learning, asymptotic analysis, concentration inequalities, uniform convergence, Rademacher complexity, and nonconvex optimization.
- Delivered presentations on Generalization Bounds via Uniform Convergence, Rademacher Complexity Bounds, and the Implicit Regularization Effect.

AWARDS & HONORS

- *Third Prize*, **Chinese Mathematics Competitions**, 2024
- *Second Prize*, **National College Students Biomedical Engineering Innovation Design Competition**, 2023

SKILLS

- **Programming Languages:** Python (Pytorch), Java, R, Matlab.
- **Software & Tools:** Texmacs, Latex, Git, Hadoop, Spark, Yarn, Linux Shell Script, Markdown.
- **Language:** Mandarin (native), English (proficient, IELTS: 7.5/9)